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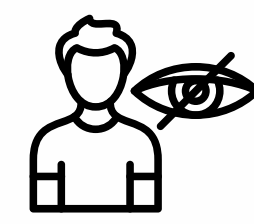
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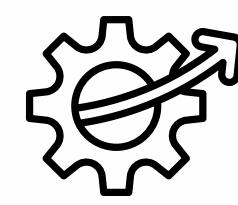
Abstract

Munir is an AI-powered assistive smart glasses system designed to support visually impaired users with real-time awareness of their surroundings. By leveraging computer vision and artificial intelligence, Munir identifies people instantly through face recognition, reads text aloud, and accurately identifies colors and currency on the spot. The system further enhances perception through smart object detection and detailed scene description, delivering instant audio feedback. Additionally, Munir includes an integrated reminders feature and an SOS messaging system for emergency alerts, promoting independence and safety in daily life — reflecting the potential of intelligent technology to improve the quality of life for individuals with visual impairments.

Objectives



Address the Needs of Visually Impaired Individuals: Enable real-time recognition and audio feedback for visually impaired users.



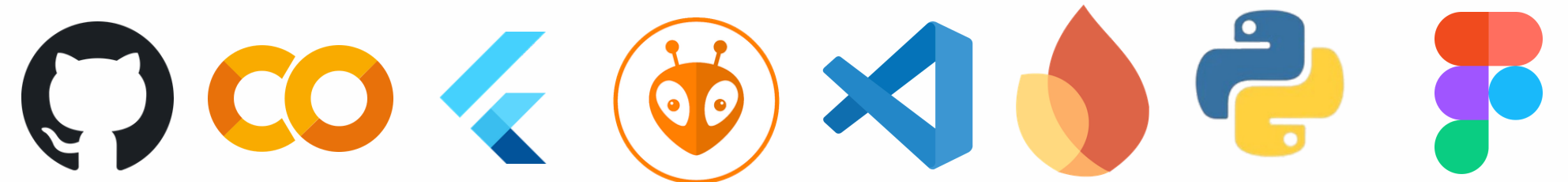
Promote Autonomy and Quality of Life:

Empower users to manage daily routines independently while enhancing confidence and overall well-being.



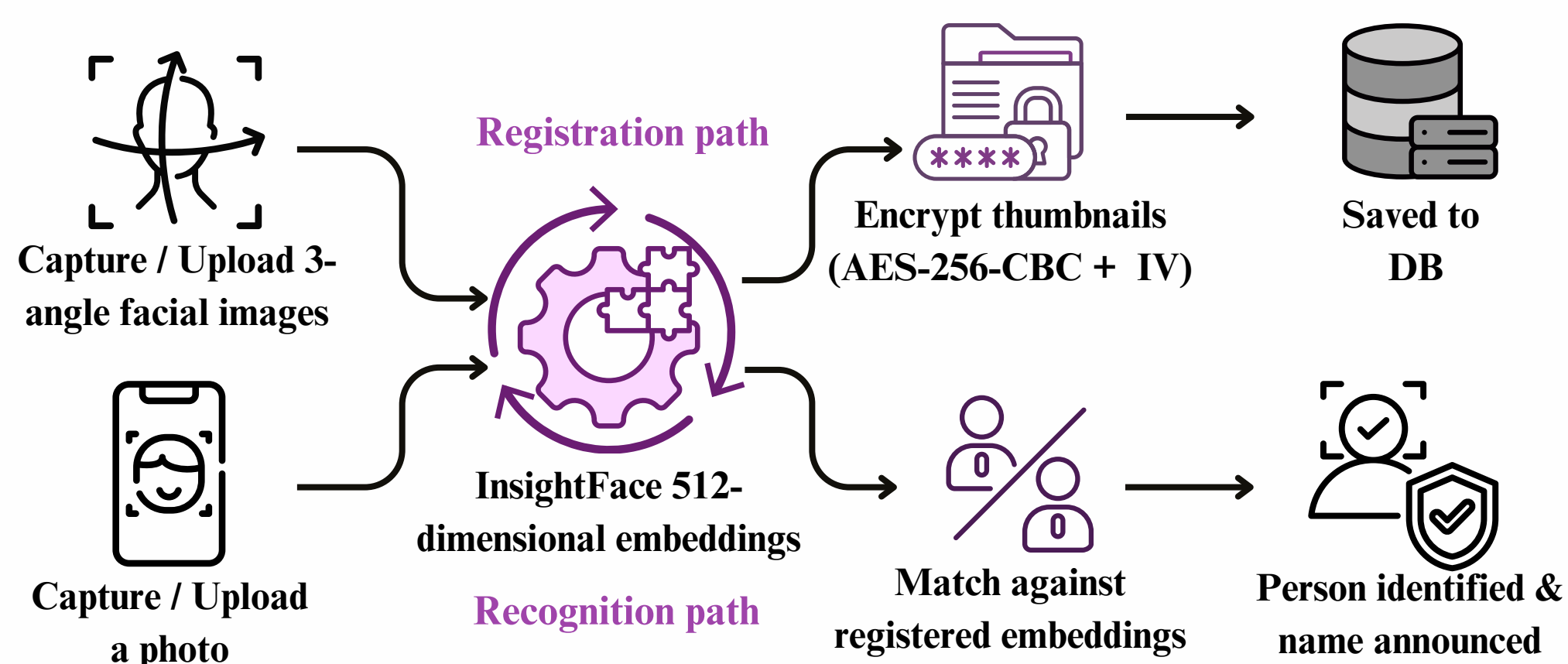
Ensure Safety and Emergency Support: Provide proactive alerts and an SOS messaging system to ensure user safety in critical situation

Tools

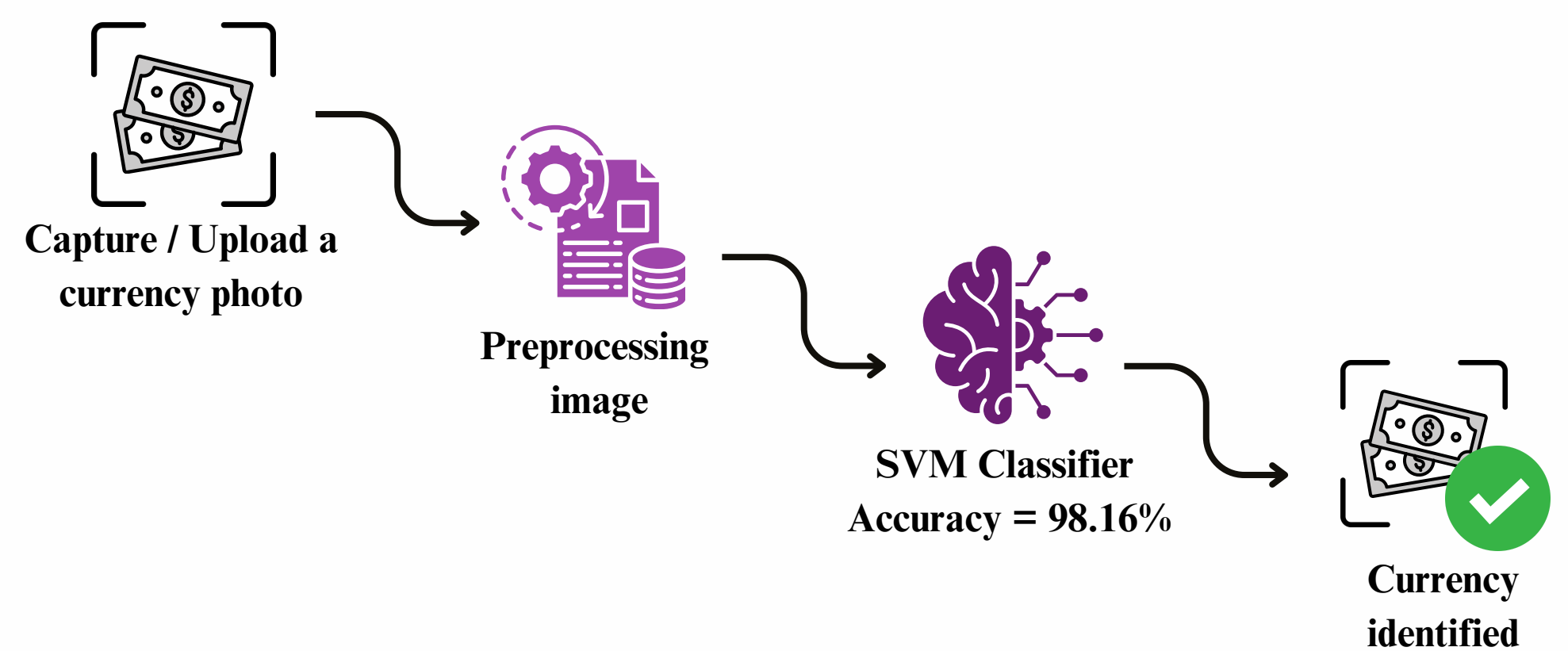


AI Models

Face Recognition



Currency Identification



System



Bluetooth
(BLE v5.3)

Microphone Built-in I2S Mic

Camera XIAO
ESP32 S3 Sense

